BRUTE FORCE ATTACK DETECTION USING WINDOWS SECURITY LOGS

SOC ANALYST PROJECT

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(B.Tech in Cyber Security 2026)

ABSTRACT

This project focuses on detecting a brute force attack using Windows Security Event Logs. A brute force attack was simulated by performing multiple failed login attempts on a system. These attempts generated security logs, which were then analyzed using the Windows Event Viewer.

The analysis focused on identifying repeated failed login attempts using specific Event IDs. A pattern of multiple failed logins within a short time interval was observed, which is a common indicator of a brute force attack.

This project demonstrates how a SOC Analyst can monitor and analyze system logs to detect potential security threats and take appropriate actions to prevent unauthorized access.

OBJECTIVE

- To simulate a brute force attack on a system
- To analyze Windows Security Event Logs
- To identify failed login attempts using Event ID 4625
- To detect suspicious login patterns
- To understand the role of a SOC Analyst in threat detection

WHAT IS BRUTE FORCE ATTACK?

A brute force attack is a method used by attackers to gain unauthorized access to a system by repeatedly trying different passwords until the correct one is found.

CHARACTERISTICS

- Multiple failed login attempts
- Repeated access attempts in short time
- Targeting a specific user account

EXAMPLE

An attacker tries multiple passwords on a user account continuously until access is granted.

METHODOLOGY

Step 1: Attack Simulation
The system was locked and multiple incorrect passwords were entered to simulate a brute force attack.

Step 2: Log Collection
The Windows Event Viewer was opened to collect system security logs.

Step 3: Log Navigation
Navigated to:
Windows Logs → Security

Step 4: Log Filtering
Logs were filtered using Event ID 4625 to identify failed login attempts.

Step 5: Data Analysis
The logs were analyzed to identify repeated failed login attempts and detect suspicious patterns.

WINDOWS LOG ANALYSIS

Windows Security Logs were analyzed using Event Viewer to identify login activities

IMPORTANT EVENT IDS:

- Event ID 4625 → Failed Login Attempt
- Event ID 4624 → Successful Login

LOG DETAILS OBSERVED:

- Username of the account
- Time of login attempts
- Frequency of failed logins
- Logon type

DETECTION LOGIC

A brute force attack is identified by analyzing patterns in login attempts.

DETECTION CONDITIONS:

- Multiple failed login attempts (Event ID 4625)
- Attempts occur within a short time period
- Same user account is repeatedly targeted

ANALYSIS

The logs showed a high number of failed login attempts within a short duration, indicating abnormal behavior compared to normal login activity.

NORMAL VS ATTACK BEHAVIOR

NORMAL BEHAVIOR

- Few login attempts
- Successful login within 1–2 tries
- No repeated failures
- Normal time intervals between attempts

ATTACK BEHAVIOR

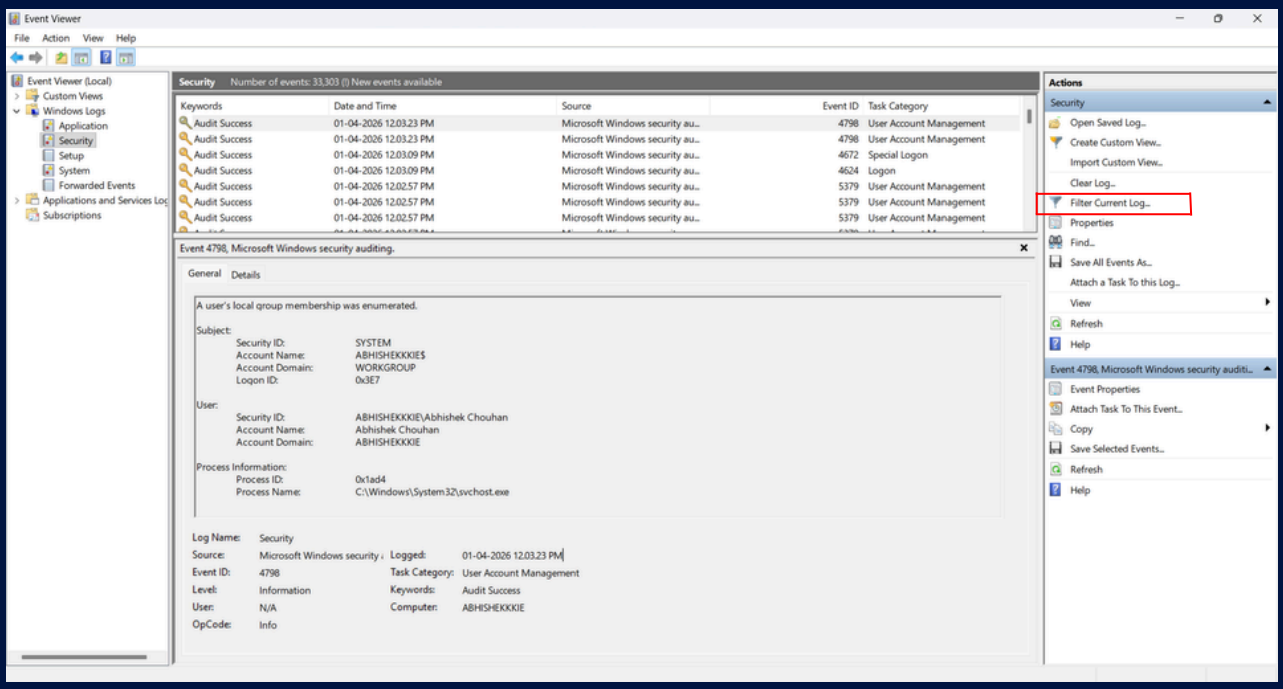
- Multiple failed login attempts
- Continuous login failures
- Attempts occur in a short time period
- Same account is repeatedly targeted

COMPARISON

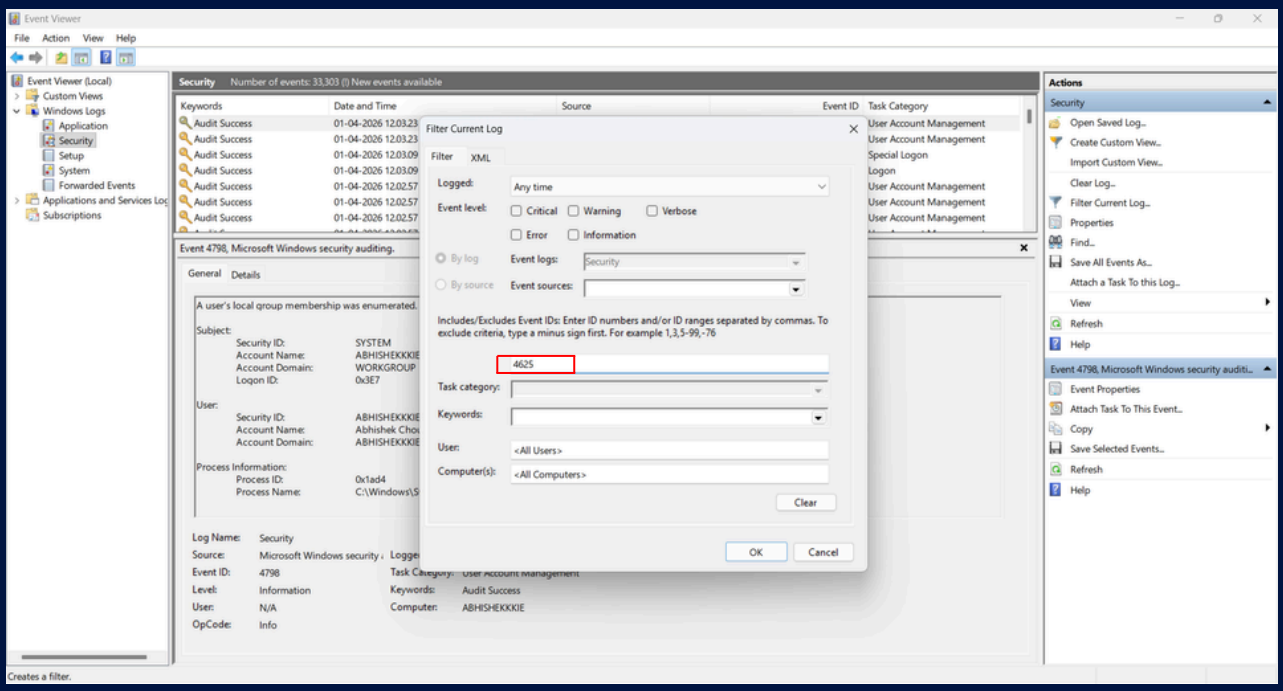
Normal user activity is limited and successful, whereas brute force attacks show repeated and rapid failed login attempts

EVIDENCE AND LOG ANALYSIS

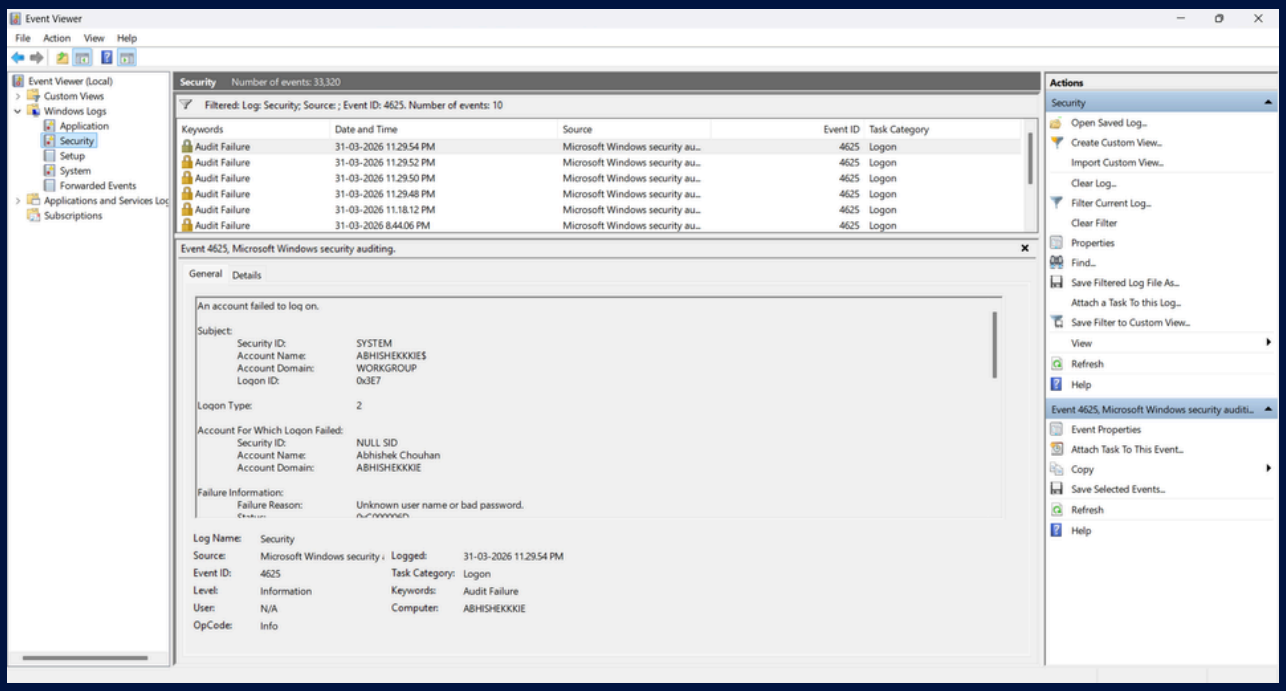
Windows Event Viewer showing Security logs with “Filter Current Log” option used to analyze login activity



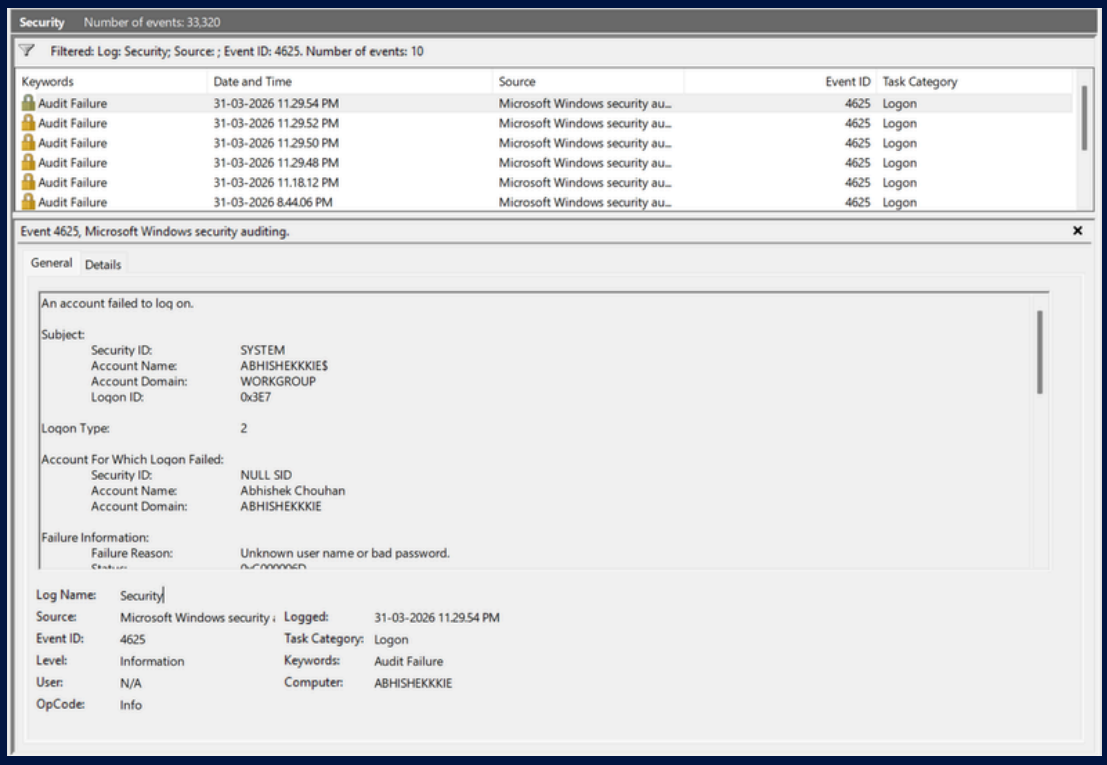
The “Filter Current Log” option is used to filter security logs by Event ID 4625, which helps in identifying multiple failed login attempts during analysis.



The screenshot below shows multiple failed login attempts recorded in Windows Security Logs.
Each failed login attempt is identified by Event ID 4625, which indicates unsuccessful authentication.



- Event ID: 4625 (Failed Login)
- Username: ABHISHEKKKIE
- Timestamp: Shows repeated login attempts within a short time period



IMPACT OF ATTACK

A brute force attack can have serious consequences if successful.

POSSIBLE IMPACTS

- Unauthorized access to user accounts
- Compromise of sensitive information
- Increased risk of system breach
- Disruption of normal system operations

SECURITY RISK

If the attacker successfully guesses the password, they can gain full access to the system and perform malicious activities.

MITIGATION STRATEGIES

To prevent brute force attacks, the following security measures can be implemented :

- Enable account lockout policy after multiple failed login attempts
- Use strong and complex passwords
- Implement Multi-Factor Authentication (MFA)
- Monitor login activity regularly
- Configure alerts using SIEM tools

ROLE OF A SOC ANALYST

A Security Operations Center (SOC) Analyst is responsible for monitoring, detecting, and responding to security threats in an organization.

- Monitor security logs and alerts
- Identify suspicious activities
- Investigate potential security incidents
- Analyze attack patterns
- Respond to and mitigate threats

CONCLUSION

This project successfully demonstrated the detection of a brute force attack using Windows Security Event Logs. By simulating multiple failed login attempts and analyzing Event ID 4625, a clear pattern of suspicious activity was identified.

The project highlights the importance of log monitoring and analysis in identifying security threats. It also demonstrates how a SOC Analyst can detect and respond to potential attacks using basic tools like Event Viewer.